<u>standing wave</u>: obtained by taking superposition of two oppositely traveling waves.

$$y_R(x,t) = y_{max} \sin(kx - \omega t)$$

$$y_L(x,t) = y_{max} \sin(kx + \omega t)$$

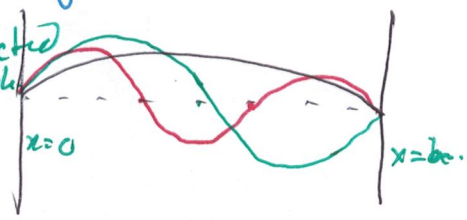
$$y(x,t) = y_L + y_R = 2 y_{max} \sin(kx) \cos(\omega t).$$

$\longrightarrow$ Standing wave!

"Derivation": R-J formula

Rakeigh considered the radiation field as a collection of normal modes in a cubical cavity of size L (Black Body)

Since the radiation is confined, the modes are standing wave which have nodes on the walls of the cavity. The permitted wave vecors are. (The idea is: not all modes of radiation (λ/ν) possible, but selected ones possible)



$$k_x = \frac{2\pi}{\lambda_x} = \frac{2\pi}{(2L/n_x)} = \frac{n_x \pi}{L}$$

Similarly $k_y = \frac{n_y \pi}{L}$, $k_z = \frac{n_z \pi}{L}$ here, $n_x, n_y, n_z \equiv 1, 2, 3 \ldots$

Defining $\vec{n} = n_x \hat{i} + n_y \hat{j} + n_z \hat{k}$, we have $k^2 = \frac{4\pi^2 \nu^2}{c^2} = \frac{\pi^2 n^2}{L^2}$

$$\Rightarrow \quad n^2 = \frac{4\pi^2 \nu^2}{c^2} \qquad \boxed{n_x^2 + n_y^2 + n_z^2 = \frac{4\pi^2 \nu^2}{c^2}} \qquad k = \frac{2\pi}{\lambda} = \frac{2\pi \nu}{c}$$

$\frac{c}{\lambda} = \nu \Rightarrow \frac{1}{\lambda} = \frac{\nu}{c}$

This last eq^n is an equation of a sphere of radius $\frac{2\pi\nu}{c}$

(*) Each distinct set of positive integer values of n_x, n_y, n_z represent one mode of vibration. (all allowed sets of n in 1st octant) — These are allowed modes of radiation.

how many such modes possible?

(*) Since the +ve values of n_x, n_y, n_z lie in one octant, the number of modes in the frequency interval ν and $\nu + d\nu$ is the volume of k-space in the interval of k and k+dk, divided by $(\pi/L)^3$ which is the volume occupied by a single point in k-space.

To account for two polarization states of a standing wave modes (geometrical orientation of oscillation)

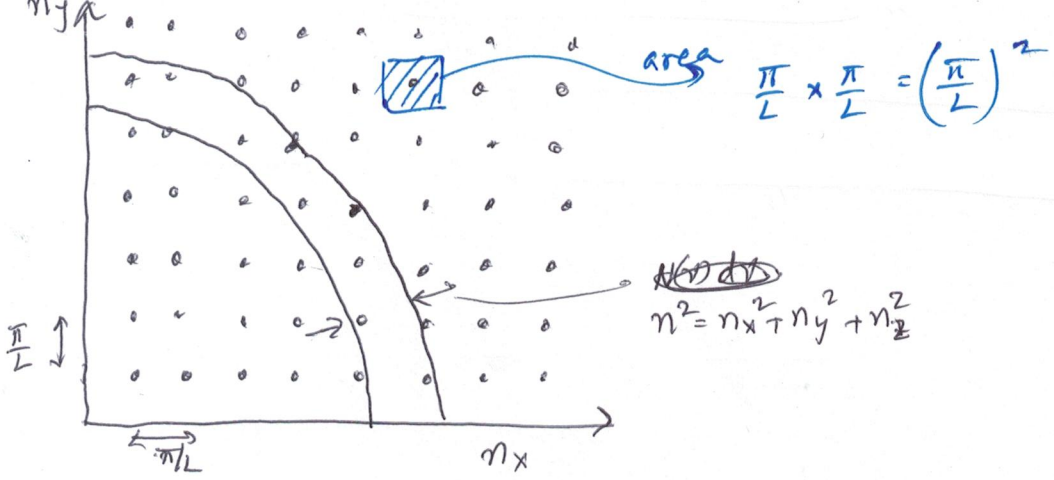
i.e. $$N(\nu)\, d\nu = \frac{1}{8} \cdot 2 \cdot \frac{4\pi k^2 dk}{(\pi/L)^3}$$

$$= \frac{1}{4} \cdot \left(\frac{L}{\pi}\right)^3 \times 4\pi \cdot \frac{(2\pi)^3}{c^3} \nu^2 d\nu$$

$$= \frac{8\pi L^3 \nu^2 d\nu}{c^3}$$

used, $k^2 = \frac{4\pi^2}{c^2}\nu^2$

$\Rightarrow$ $2k\,dk = \frac{4\pi^2}{c^2} 2\nu\, d\nu$



Next we consider the mean energy of each mode of vibration at temperature T be $\boxed{kT}$. (comes from equipartition theorem)

Then, the energy density of radiation in the frequency interval of ν & $\nu + d\nu$ per unit volume

i.e. $\boxed{\text{(The number of modes)}} \times \boxed{\text{(average energy per mode)}}$

$$\Rightarrow \boxed{E(\nu, T)\, d\nu = \frac{8\pi k_B T}{c^3} \nu^2 d\nu}$$

Raleigh-Jean's law

Equipartition theorem:
The total energy in thermal equilibrium is equally distributed among all its degree of freedom (here, all modes)

Planck's law:

Radical assumption: Some agent (called oscillator) in the cavity walls could emit or absorb energy only in chunks (in quanta), and the chunk size is proportional to the frequency of the oscillator.

Consider in a brief interval of time ($8h\nu$) of energy becomes available to be absorbed or to be emitted. There are variety of ways for this to happen, let us consider a few.

- 8 oscillators at fundamental frequency ν could each absorb or emit 1 quantum of energy.

$$\therefore \underbrace{8}_{\#\text{ of oscillator}} \times \underbrace{1h\nu}_{\text{each of quantum}} = 8h\nu \rightarrow \text{total energy.}$$

⊙ Another possibility could be

$$4 \times (1h\nu) + 2 \times (2h\nu) = 8h\nu.$$

each of these two oscillator carries quantum of energy corresponds to freq. 2ν (1st harmonic)

⊙ Another

$$3 \times (1h\nu) \quad 1 \times (3h\nu) + 1 \times (2h\nu) = 8h\nu.$$

and so on.

Thus, the available total energy can be allocated to oscillators in many ways. And oscillators can appear with different ~~quantum size~~, freq., but multiples of the fundamental one.

~~Planck wanted to calculate the average energy per~~ mo

⊛ Planck used the calculation of the number of modes of radiation within the frequency interval of ν & $\nu + d\nu$, developed earlier by Raleigh-Jeans.

⊛ But he used a different calculation to estimate the average energy per mode, as we describe below.

Planck used Maxwell-Boltzmann statistics to decide the number of oscillator available with energy $(nh\nu)$, given by,

$$N_n = A \exp\left[-\frac{nh\nu}{k_B T}\right]$$

A is a normalization constant

The total energy of oscillators having energy $(nh\nu)$ is

$$E_n = N_n (nh\nu)$$

Hence, the total energy of all oscillators in the blackbody : $E = \sum^{\infty} E_n = \sum^{\infty} N_n (nh\nu)$

The total number of oscillators in the Blackbody is,

$$N = \sum_n N_n$$

Hence, the average energy of the oscillator is:

$$\langle E \rangle = \frac{\text{total energy of the oscillator}}{\text{total no. of oscillator.}}$$

$$= \frac{\sum_{n=0}^{\infty} N_n (nh\nu)}{\sum_{n=0}^{\infty} N_n} = \frac{\sum_{n=0}^{\infty} nh\nu\, e^{-\frac{nh\nu}{k_B T}}}{\sum_{n=0}^{\infty} \exp\left[-\frac{nh\nu}{k_B T}\right]}$$

Let $x = \frac{h\nu}{k_B T}$

$$\Rightarrow \langle E \rangle = k_B T \frac{\sum n x e^{-nx}}{\sum e^{-nx}} = \frac{-x k_B T \frac{d}{dx}\left(\sum_{n=0}^{\infty} e^{-nx}\right)}{\sum e^{-nx}}$$

$$\boxed{\sum_{n=0}^{\infty} e^{-nx} = \frac{1}{1-e^{-x}}}, \quad \frac{d}{dx}\left[\sum_{n=0}^{\infty} e^{-nx}\right] = -\frac{e^{-x}}{(1-e^{-x})^2}$$

$$\Rightarrow \langle E \rangle = \frac{x k_B T}{(e^{x}-1)}$$

$$\Rightarrow \langle E \rangle = \frac{h\nu}{(e^{h\nu/k_B T}-1)} = \frac{hc/\lambda}{(e^{hc/\lambda k_B T}-1)}$$

This is the avg. energy instead of $(k_B T)$ in R-J form.

Finally,

$$\boxed{E(\nu,T)\,d\nu = \frac{8\pi h}{c^3} \frac{\nu^3 d\nu}{(e^{h\nu/k_B T}-1)}}$$

and,

$$\boxed{E(\lambda,T)\,d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{(e^{hc/\lambda k_B T}-1)}}$$